UPchange: Unpaired Supervised Semantic Change Detection for High-Resolution Remote Sensing Imagery

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***Abstract*—High-resolution remote sensing imagery semantic change detection (SCD) identifies both the regions of change and land cover variations in multitemporal data, which has huge application potential in many fields. However, most of the existing methods rely on bitemporal remote sensing images with paired change and semantic labels, which are costly and labor-intensive to acquire. In this article, we propose a novel framework named UPchange, which enables training of a semantic change detector using single-temporal labeled data. First, we generate a substantial volume of supervised semantic change labels on unpaired data by treating different semantic annotations in these data as indicators of change. Furthermore, due to the imbalance between generated positive change samples and negative unchanged samples, a new formulation named ChangeType was proposed to transform the SCD problem into a classification problem, in which each type of change is assigned a specific class, thus splitting the dominant binary change samples into different semantic change samples. Lastly, to bridge the domain gap between unpaired data with pseudo-labels in the training process and paired data in the inference process, a temporal change adaption (TCA) was used, with unpaired data as the source domain and paired data as the target domain. We conducted extensive experiments on two open-source SCD datasets (Hi-UCD mini and SECOND). The proposed UPchange framework outperforms two benchmarks under single-temporal settings and can achieve superior performance under bitemporal settings, demonstrating the efficiency and superiority of the method. Moreover, we deployed the model on three large-scale regions for semantic change inference, demonstrating its generalizability and robustness in real-world applications. Code is available at:https://github.com/xian0706/UPchange**

***Index Terms*—Deep learning, semantic change detection (SCD), unsupervised domain adaptation.**

I. Introduction

HIGH-RESOLUTION remote sensing imagery change detection, an efficient way of observing the world, identifies differences in the state of an object or phenomenon by monitoring it at different times [1], has great potential in urban planning [2], [3], agriculture monitoring [4], disaster assessment [5], [6] and many others. Most of the existing work focuses on binary change detection (BCD) [7], [8], which only displays the regions of change. Semantic change detection (SCD) seeks to capture "from-to" change information [9]. It not only provides binary change information but also identifies semantic land cover at different times, making it a more challenging task.

The primary objective of SCD is to efficiently and rapidly identify differences in land cover between remote sensing images of the same area taken at different times. A direct approach to change detection is to compare multitemporal remote sensing data pixel by pixel. Change vector analysis (CVA) methods [10], [11], which construct a change vector by combining the image vectors of corresponding pixels from two different images, attempt to classify semantic changes from this perspective. Some object-based methods [12], [13] consider the spatial relationships between pixels, tackling the SCD task in two steps: object extraction and comparison. However, all of these methods are based on hand-crafted representations developed by experts.

Due to its strong feature extraction capabilities and robustness, deep convolutional neural network (CNN) [14] has been widely used to solve change detection (including binary and segmentation) tasks [15], [16]. Almost all current SCD methods are based on the Siamese multitask network [9], which consists of a dual share-weight semantic encoder that extracts semantic features from bitemporal remote sensing images, a semantic decoder that outputs the land cover result of temporal data, and a change detector that identifies the binary change regions. Though many works based on this framework achieved significant accomplishments, they all share a major drawback that requires pairwise labeled bitemporal images, which are very expensive and labor-consuming. Some works aim to address the problem of the need for a large number of pairwise labeled bitemporal images. A semisupervised learning method has proven to be an efficient way for change detection, where a change detector is first trained on a small set of paired, labeled data and then fine-tuned on a large amount of unlabeled data using techniques such as consistency across different image augmentations [17], similar change feature between labeled and unlabeled

Received 13 June 2025; revised 30 November 2025 and 8 March 2026; accepted 26 May 2026. Date of publication 17 June 2026; date of current version 29 June 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFC3015403 and in part by the National Natural Science Foundation of China under Grant 42171336. *(Corresponding author: Ailong Ma.)*

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Digital Object Identifier 10.1109/TGRS.2026.3704148

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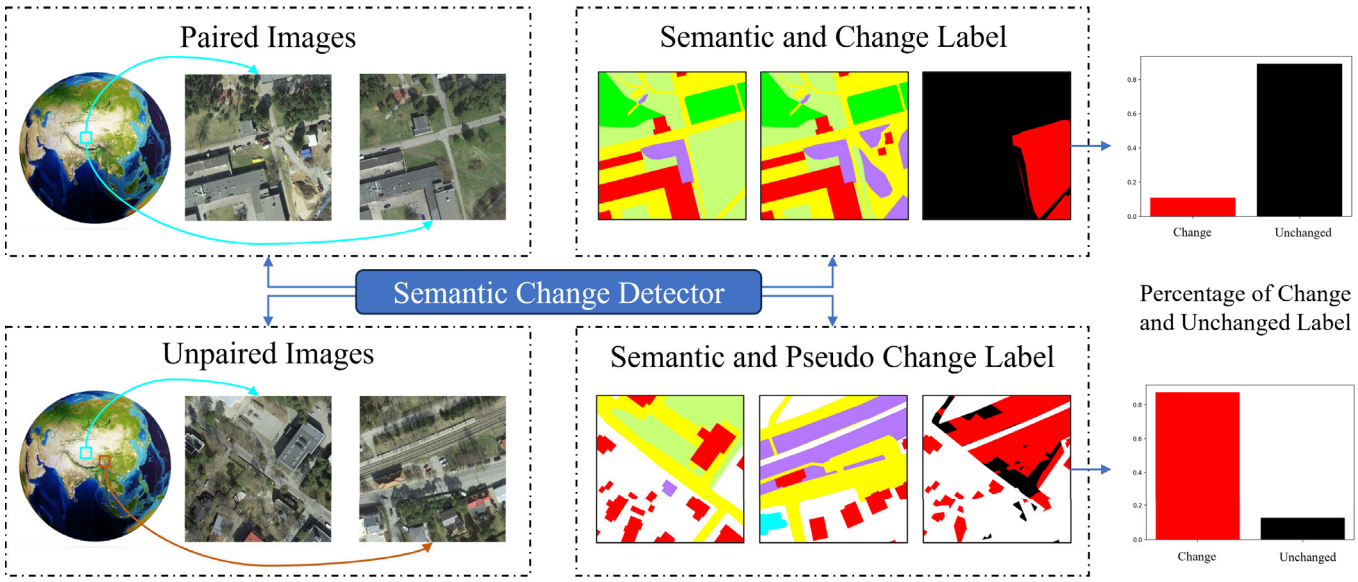


Fig. 1. Comparison of paired and unpaired SCD. There is a huge difference in the percentage of changed and unchanged labels in the two frameworks.

data [18], or other strategies. However, these methods still need some bitemporal paired images with their semantic annotations. Some works try to solve this problem based on single-temporal unpaired data and semantic annotations, for example, ChangeStar [19] and ChangeStar2 [20] generate pseudo-change labels from the single semantic label. Another approach to address this issue is synthetic data, which generates unreal images by land cover labels or objects from other data [21], [22], [23] using a generative model such as a generative adversarial network (GAN) or diffusion model. However, these methods do not allow end-to-end training of the change detector, and they highly depend on the quality of the data generated in the previous step. Most existing generative models are designed for single-class land cover and lack strong capability in generating multiclass scenarios; the data that are synthesized often cannot fully match real-world data.

Existing sub-meter-level land cover products hold significant potential for high-resolution semantic change prediction learning, as they provide precise and sufficient land cover information. However, this land cover information only comes from single-temporal data, which are unpaired and lack the time information needed by SCD in multitemporal applications. To solve the problem above, we proposed a new deep learning SCD framework named UPchange, which trains a semantic change detector using pseudo-semantic change labels generated by single-temporal data and annotations. An efficient formulation was proposed to solve the imbalance between positive and negative samples, and an unsupervised domain adaptation method was used to transfer the knowledge from unpaired data with pseudo-labels to paired data. The main contributions of this article are summarized as follows.

1) Pseudo-semantic change labels generated by unpaired land cover data. Typically, SCD relies on paired annotations to identify changes, with these change signals arising from differences in land cover between paired datasets. However, it is possible to train a semantic change detector using both the same and different semantic annotations coming from any data, though unpaired. As Fig. 1 shows, we generate pseudo-semantic change samples by unpaired data to address the issue of the need for a pairwise labeled bitemporal dataset in SCD. The core idea of this is to treat semantic differences at the same coordinates in unpaired images (even obtained from different locations) as a change.
2) A new formulation for SCD by modeling change types on pseudo-semantic change labels. As Fig. 1 shows, under the single-temporal unpaired setting and pseudo-semantic change labels, positive samples(change labels) will dominate the majority of the training set and play a significant role in training a change detector. This imbalance in training samples will prevent the model from learning a robust change representation when it is decoupled into semantic segmentation and change detection tasks within a multitask learning framework. Actually, an intuitive solution to the SCD task is treating it as a classification task and training a classifier to model the type of every change within a classification space $H(n^2)$, where n denotes the number of land cover classes. This is invalidation under the condition of paired multitemporal images because of the huge change space and the sparse distribution of change labels. In contrast, we can generate a large number of pseudo-semantic change labels (theoretically infinite) by the way mentioned above, which can be seen as semantic change samples in the process of training a semantic change detector.
3) Bitemporal unsupervised domain adaptation for a single-temporal semantic change detector. Due to the semantic change detector being trained on unpaired data with

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pseudo-semantic change labels, while the inference process was made on paired data, which belong to different domains. Unlike other unsupervised domain adaptation methods in the field of change detection, which eliminate the semantic representation distribution differences between bitemporal paired source and target data. We propose an unsupervised domain adaptation method that uses unpaired data with single-temporal semantic annotations as the source data and attempts to transfer the knowledge to paired bitemporal target data.

In the rest of this article, we will show our work as follows. Section II introduces both the binary and SCD tasks and the commonly unsupervised domain adaptation methods briefly. Section III proposed the UPchange method in three elements. Section IV shows extensive experimental results and a large-scale application. Our conclusions are given in Section V.

II. Related Works

A. Semantic Change Detection

Compared with remote sensing image BCD, which only identifies the regions of change, SCD is a more challenging task, providing not only the binary change mask but also the land cover information of multitemporal remote sensing data.

The intuitive way to the SCD task is the postclassification comparison methods, which train a land cover classification model on single-temporal images and then compare the semantic results on multitemporal images [24]. However, this method ignores the time consistency; classification errors can accumulate, leading to false alarms, particularly along the boundaries of objects [25]. To address the loss of time consistency, Siamese multitask learning [9] based on a fully convolutional neural network (FCN) has been proposed, which uses a weight-shared CNN to encode multitemporal remote sensing data into semantic features. A semantic decoder, consisting of transpose convolution, recovers the semantic feature into the original resolution, and a change detector is used to generate a binary change mask.

Many works have achieved impressive results in the SCD task using the Siamese multitask learning framework. ChangeMask [26] improves the structure of the Siamese multitask network by considering the causal relationship between semantic and change representation. Zhu et al. [27] introduce a hierarchical sequence of training samples strategy to balance the change and the unchanged samples. Ding et al. [28] propose a temporal consistency constraint loss to alleviate the inconsistency between the semantic and change branches. Tian et al. [29] enhance the performance of the Siamese multitask learning framework by extracting the salient area information from different temporal images. However, all of these methods rely on pairwise annotations of multitemporal remote sensing data, which are very expensive and time-consuming to acquire.

B. Single-Temporal Change Detection

Change detection tasks face the challenge of limited labeled datasets. This issue is further aggravated by the high cost of creating a change detection dataset with semantic labels.

Single-temporal change detection has been explored in recent years; these methods try to forge time information based on single-temporal data. An intuitive idea of it is image generation, which synthesizes a time series of images using semantic labels. Image generation is a field in computer vision and has been widely explored by researchers. These methods include direct generation [30], GAN [31], [32], diffusion model [33], [34]. These techniques have also been applied in change detection. Changen and Changen2 [23], [35] use GANs and diffusion models, respectively, to synthesize remote sensing images with land cover. These methods heavily rely on the quality of the data generated by the previous step, but synthesized data cannot perfectly match real-world data. Some methods also rely on additional information. For example, ChangeBridge [36] proposed a drift-asynchronous diffusion bridge that can generate post-event scenes that are both spatially and temporally coherent. ChangeClip [37] leverages visual-textual information and ControlNet to generate high-quality pseudo-change pairs from widely available single-temporal images.

Another way is to generate pseudo-change labels, for example, Self-pair [22] copies a building instance from an image and pastes these data to another image; ChangeStar [19] generates the object change label of the building by comparing semantic annotations from different locations, ChangeStar2 [20] generates a semantic change label in the same way. These methods ignore the balance of positive and negative samples.

C. Unsupervised Domain Adaptation

Unsupervised domain adaptation is an efficient way to transfer learning from one labeled source domain to an unlabeled target domain, which has been used on many tasks such as image classification, land cover classification, object detection, and so on.

The intuitive approach of UDA is a domain feature difference measure alignment method, which pulls the features of data from the source and target domains together in a tough way. Most of these methods try to maximize the mean discrepancies of features extracted by the convolutional network between the source domain and the target domain [38], [39], [40]. Recently, some works have tried to solve this task using an adversarial network and self-training.

An adversarial network is an efficient way for UDA, unlike the discriminator in the original generative adversarial network, to identify real or synthetic images [31]. An adversarial network in UDA consists of a feature extractor that tries to obtain the domain-invariant feature, and a discriminator that tries to identify the feature obtained by the extractor. Many works try to solve UDA by this method at different levels, such as image level [41], [42], feature level [43], entropy level [44], and so on.

Another efficient method for this task is self-training, which generates pseudo-labels on the target domain and then trains the model on both source and target data. Some of the works try to solve the UDA task by a curriculum learning, simply to say, an easy task like properties about the domain [45], [46] or the entropy of prediction [47] has been designed to restrict the model to learn a more robust representation of the domain.

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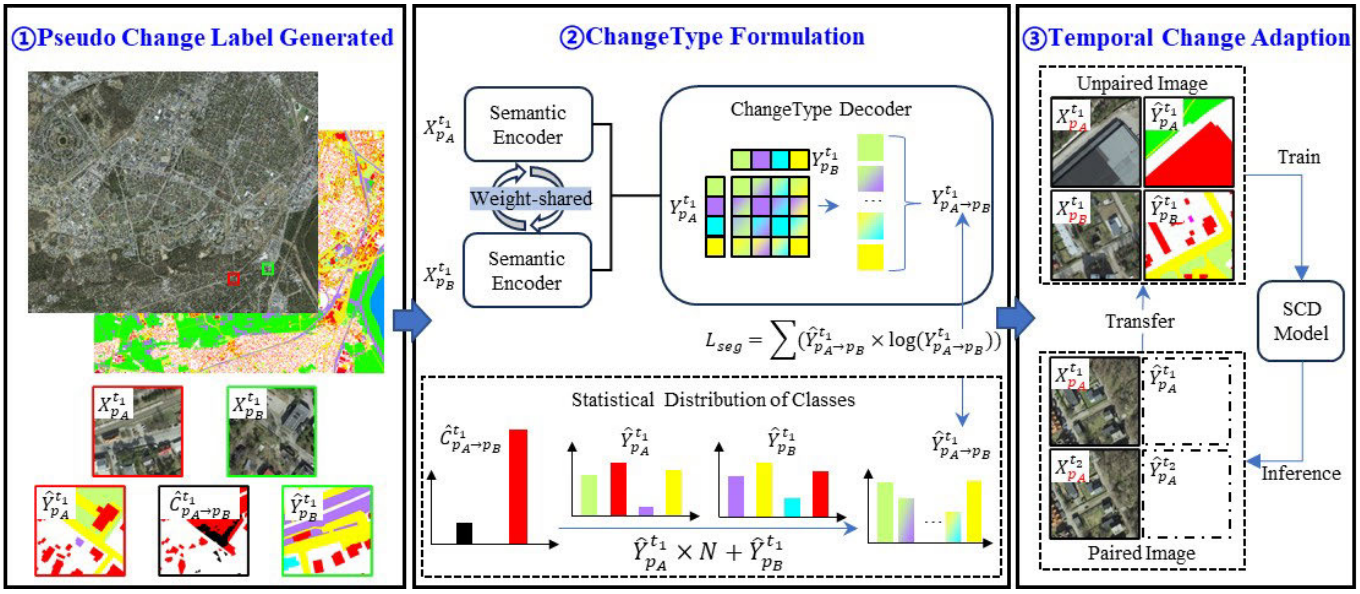


Fig. 2. Overview of UPchange. The framework comprises three components: pseudo-change labels generation by unpaired images, ChangeType formulation for imbalanced change and unchanged samples, and temporal change adaptation between unpaired and paired images.

Because UDA methods suffer from the wrong predication of unlabeled samples, so lots of the works depend on selecting reliable samples with high confidence thresholds [48], [49].

III. Methods

The overall architecture of the proposed UPchange is shown in Fig. 2. The process starts with the generation of pseudo-change labels on single-temporal data; to train an SCD detector based on this unpaired data, a formulation on Changetype was conducted. Next, the temporal change adaptation (TCA) module is applied to mitigate the domain shift between pseudo-change labels generated from unpaired single-temporal data and paired multitemporal data without annotations. The process of our algorithm can be seen at Algorithm 1. In the remainder of this section, we will give a brief introduction to our work.

A. Pseudo-Change Labels Generated for SCD Learning

The common bitemporal SCD solved by the Siamese multitask network can be factorized as [19]

$$\min_{\theta} \mathcal{L}\left(F_\theta\left(X_{p_A}^{t_1}, X_{p_A}^{t_2}\right), \hat{C}_{p_A}^{t_1\to t_2}, \hat{Y}_{p_A}^{t_1}, \hat{Y}_{p_A}^{t_2}\right) \tag{1}$$

where F and θ indicate the model and its parameter, the multitemporal paired images $X_{p_A}^{t_1}, X_{p_A}^{t_2} \in R^{N*C*H*W}$ are input into model above and get three results: the land cover prediction of bitemporal data Y^{t_1}, Y^{t_2} and a binary change mask $C_{p_A}^{t_1\to t_2} \in R^{H*W}$, the pixel (i, j) of binary change mask $C_{p_A}^{t_1\to t_2}(i, j) \in [0, 1]$ indicates its probability of being predicted to change. The process of training optimizes the model's parameters θ to minimize the difference between three predictions and their labels. This formulation implicitly assumes that the remote sensing data were obtained from the same spatial position, which is a strong assumption, as local geometric offsets, radiance differences, and other factors can disturb this condition.

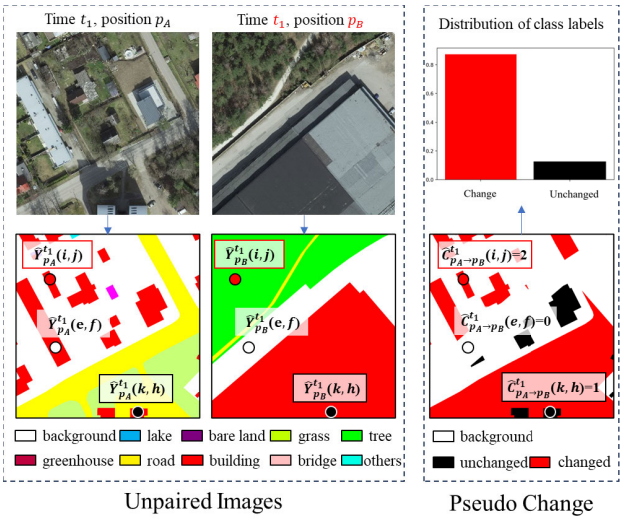


Fig. 3. Process of generating the pseudo-change label, a change label pixel is identified when two corresponding pixels possess different semantic labels.

This is one of the reasons why change detection datasets are expensive and labor-intensive to create.

In this work, this condition can be relaxed by using data obtained from different spatial positions regardless of whether their timestamps are the same. As Fig. 3 shows, the core idea is to utilize the different semantic labels between two images in different positions, a change label pixel is identified when two corresponding pixels possess different semantic labels. With the releasing of the same spatial position condition, (1) can simplified into

$$\min_{\theta} \mathcal{L}\left(F_\theta\left(X_{p_A}^{t_1}, X_{p_B}^{t_1}\right), \hat{C}_{p_A\to p_B}^{t_1}, \hat{Y}_{p_A}^{t_1}, \hat{Y}_{p_B}^{t_1}\right) \tag{2}$$

where $X_{p_A}^{t_1}, X_{p_B}^{t_1} \in R^{N*C*H*W}$ were unpaired data that can be got from single-temporal, $\hat{Y}_{p_A}^{t_1}, \hat{Y}_{p_B}^{t_1}$ were their semantic label and $\hat{C}_{p_A\to p_B}^{t_1} \in R^{N*H*W}$ was a pseudo-change label generated with

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Algorithm 1 Process of UPchange

Input: Source: $(X^{t_1}_{p_A}, X^{t_1}_{p_B})$, Target: $(X^{t_1}_{p_A}, X^{t_2}_{p_A})$, Learning rate: γ, Warm up iterations: I^w, Generate pseudo-label iterations: I^g, Total iterations: I^t,

Initialize model parameters θ
Pseudo-Change Label Generated
if $\hat{Y}^{t_1}_{p_A} == \hat{Y}^{t_1}_{p_B}$ **then**
  $\hat{C}^{t_1}_{p_A \to p_B} = 0$
else
  $\hat{C}^{t_1}_{p_A \to p_B} = 1$
end if
ChangeType Formulation
$\hat{Y}^{t_1}_{p_A \to p_B} = \hat{Y}^{t_1}_{p_A} \times N + \hat{Y}^{t_1}_{p_B}$
for Iters = 1 to I^w **do**
  $Y^{t_I}_{p_A \to p_B} = \mathcal{F}_\theta(X^{t_1}_{p_A}, X^{t_1}_{p_B})$
  $\mathcal{L}_s = BCELoss(\hat{Y}^{t_1}_{p_A \to p_B}, Y^{t_I}_{p_A \to p_B})$
  $\theta = \theta + \nabla_\theta \mathcal{L}_s$
end for
Temporal change adaptation
for Iters = I^w to I^t **do**
  if mod(Iters, I^g) = 0 **then**
    Pseudo-labels generate
    $\hat{Y}^{t_1 \to t_2}_{p_A} = \mathcal{F}_\theta(X^{t_1}_{p_A}, X^{t_2}_{p_A})$
    $\hat{Y}^{t_1 \to t_2}_{p_A} = \mathcal{F}^{selection}(\hat{Y}^{t_1 \to t_2}_{p_A})$
  end if
  $Y^{t_I}_{p_A \to p_B} = \mathcal{F}_\theta(X^{t_1}_{p_A}, X^{t_1}_{p_B})$
  $\hat{Y}^{t_1 \to t_2}_{p_A} = \mathcal{F}_\theta(X^{t_1}_{p_A}, X^{t_2}_{p_A})$
  $\mathcal{L}_s = BCELoss(\hat{Y}^{t_1}_{p_A \to p_B}, Y^{t_I}_{p_A \to p_B})$
  $\mathcal{L}_t = BCELoss(\hat{Y}^{t_1 \to t_2}_{p_A}, Y^{t_I \to t_2}_{p_A})$
  $\theta = \theta + \nabla_\theta \mathcal{L}_s + \nabla_\theta \mathcal{L}_t$
end for

XOR of semantic label $\hat{Y}^{t_1}, \hat{Y}^{t'_1}$. As Fig. 3 shows, the value of the pseudo-change label can be calculated by

$$\hat{C}^{t_1}_{p_A \to p_B}(i,j) = \begin{cases} 0, & if\ \hat{Y}^{t_1}_{p_A}(i,j) = 0\ or \\ & \hat{Y}^{t_1}_{p_B}(i,j) = 0 \\ 1, & if\ \hat{Y}^{t_1}_{p_A}(i,j) = \hat{Y}^{t_1}_{p_B}(i,j)\ and \\ & \hat{Y}^{t_1}_{p_A}(i,j) \neq 0 \\ 2, & if\ \hat{Y}^{t_1}_{p_A}(i,j) \neq \hat{Y}^{t_1}_{p_B}(i,j)\ and \\ & \hat{Y}^{t_1}_{p_A}(i,j) * \hat{Y}^{t_1}_{p_B}(i,j) \neq 0 \end{cases} \tag{3}$$

where 0 means background, 1 means unchanged, 2 means change, $\hat{C}^{t_1}_{p_A \to p_B}$ means the binary pseudo-change map, and $\hat{Y}^{t_1}_{p_A}$ means the semantic label. The background portion within the binary pseudo-change map (white color) arises because the corresponding pixels in the first or second labels are background.

Using the method above, we construct a basic SCD sample with only unpaired semantic labels. However, these pseudo-labels mostly consist of change labels due to the significant semantic differences between the unpaired data. To balance the distribution of positive and negative samples, some self-pair data were introduced during the process of training. These data are compared to themselves through image transformations, ensuring that there are no change labels in these samples.

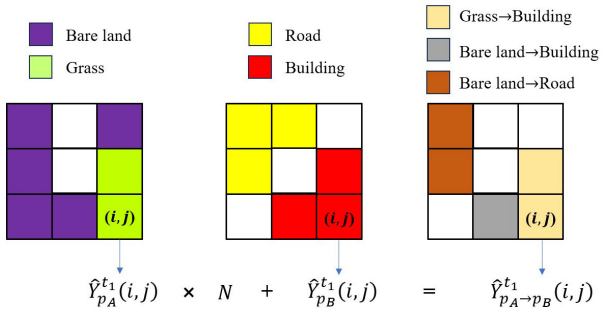


Fig. 4. ChangeType Formulation. The data used for training the model is obtained by multiplying the semantic label class value of the first label by the total number of semantic categories N, and then adding the corresponding value of the second label.

B. ChangeType Formulation Based on Pseudo-Change Labels

Though we add some self-pair unchanged labels to balance the imbalance between negative and positive samples, positive samples still dominate a large portion of the training set due to the significant semantic differences in unpaired single-temporal data. Meanwhile, generalizing a large number of different pseudo-change types to a semantic-agnostic class will confuse the model and hinder its ability to learn the change representations from real data.

Rethink the SCD task based on paired data; an intuitive solution is direct-CD, where all types of changes are modeled. However, due to the sparsity of change labels in real paired multitemporal remote sensing data, this huge classification space $H(n^2)$, where n denotes the number of land cover classes, makes the classifier difficult to train because of the limited number of positive samples. In contrast, when we transfer to the question raised above, the drastic semantic change between unpaired data brings a large number of change labels, providing sufficient positive samples for the model to learn class-aware change representations, at least in theory. Meanwhile, the Changetype formulation transforms the SCD task into a classification task, in which the pseudo-changed labels can be segmented into different semantic change labels (from bare land to building, etc.). Through this method, we have decomposed the dominant pseudo-change labels into multiple categories of semantic change labels. Although the issue of class imbalance persists, it is not severe and can be effectively addressed by many existing semantic segmentation methods. Unlike the most popular Siamese multitask framework, which decouples the SCD task into a semantic branch (for land cover classification) and a change branch (for class-agnostic change detection), we adopt the ChangeType modeling to learn semantic-aware changes and simplify (2)

$$\min_\theta \mathcal{L}\left(F_\theta\left(X^{t_I}_{p_A}, X^{t_I}_{p_B}\right), \hat{Y}^{t_1}_{p_A \to p_B}\right) \tag{4}$$

in which $\hat{Y}^{t_1}_{p_A \to p_B}(i,j)$ can be calculated by

$$\hat{Y}^{t_1}_{p_A \to p_B}(i,j) = \begin{cases} 0, & if\ \hat{Y}^{t_1}_{p_A}(i,j) * \hat{Y}^{t_1}_{p_B}(i,j) = 0 \\ \hat{Y}^{t_1}_{p_A}(i,j) * N + \hat{Y}^{t_1}_{p_B}(i,j), & else \end{cases} \tag{5}$$

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where $\hat{Y}^{t_1}_{p_A \to p_B} \in R^{N*C^2*H*W}$ and N is the number of land cover classes, as Fig. 4. The structure of multitask learning is based on bitemporal paired data, and our change type modeling is based on single-temporal unpaired data.

The network architecture of our work consists of a semantic-aware encoder, a U-net semantic change detector, and an optional semantic decoder. The proposed framework represents a deep learning strategy for the SCD task under the setting of unpaired semantic labels, allowing any SCD network to be incorporated into this framework. The semantic branch is optional because the semantic change branch also provides the land cover results. We will discuss the benefit brought by this branch in Section IV. The loss function for this modeling is the commonly used multiclass cross-entropy

$$L_{seg} = -\sum_{i=0}^{n} \hat{Y}^{t_1}_{p_A \to p_B} \log\left(Y^{t_1}_{p_A \to p_B}\right) \tag{6}$$

in which y_i is the one-hot label of semantic change, and p_i is the SoftMax output of the semantic change detector.

C. Temporal Domain Adaptation for Semantic Change Detector

Using the method described above, we construct a pseudo-SCD sample and train a semantic change detector. However, the real changed and unchanged features between paired multitemporal data differ from our pseudo-labels. This discrepancy arises from the domain shift between paired and unpaired data, so a self-training adaptation method named TCA was proposed to solve this problem. Unlike other UDA methods in the change detection field that require both source and target bitemporal paired images to align the semantic change feature distribution differences between source and target domains, UPchange only requires unpaired source images and paired target images.

Self-training is an important category of methods in the field of unsupervised domain adaptation. These approaches begin by using a model which trained on the source domain to infer the target domain data, then filter the inference results and obtain the most accurate pseudo-labels as possible. These filtered pseudo-labels are subsequently treated as ground truth to train the model on the target domain data. Through multiple iterations of inference, filtering, and training, the model is progressively adapted from the source domain to the target domain. Regarding the threshold for pseudo-label selection, a conventional approach typically employs a fixed threshold or percentage to select the top confidence samples, which is not ideal for our method. This is because the imbalance in semantic labels may be exacerbated due to the random sampling. So we propose a dynamic strategy that combines both the threshold and percentage to select the most robust samples

$$Y^{t_l \to t_2}_{p_A}(i,j,k) = \begin{cases} 1, & if\ k = \arg\max_k \hat{Y}^{t_1 \to t_2}_{p_A}(i,j,k) \\ & and\ T_k < \max_k \hat{Y}^{t_1 \to t_2}_{p_A}(i,j,k) \\ & and\ P_k < \max_k \hat{Y}^{t_1 \to t_2}_{p_A}(i,j,k) \\ 0, & else \end{cases} \tag{7}$$

where P_k is the percentage, and the pseudo-labels will select the top $P_k\%$ samples with the highest confidence for class k to maintain class balance in the pseudo-samples. $T_k \in [0,1]$ is the confidence threshold that selects samples most likely to be correct, ensuring the quality of the pseudo-samples. Unlike the fixed P_k or T_k in CBST, we designed a dynamic linear strategy to adjust these hyperparameters during the training process. The strategy can be expressed as a function of step t and total steps T

$$P_k = P^l_k + \frac{t}{T*2} \tag{8}$$

$$T_k = 1 - \frac{t}{T*2} \tag{9}$$

P^l_k is the percentage selected at the first time to generate pseudo-labels, which was set to 0.2 in this work.

In the inference process, the result of classification $Y^{t_1 \to t_2}_{p_A}$ on bitemporal data $X^{t_1}_{p_A}$, $X^{t_2}_{p_A}$ can be transferred into two semantic map $Y^{t_1}_{p_A}$, $Y^{t_2}_{p_A}$ and a binary change map $C^{t_1 \to t_2}_{p_A}$

$$Y^{t_l}_{p_A} = Y^{t_l \to t_2}_{p_A} / N \tag{10}$$

$$Y^{t_2}_{p_A} = Y^{t_l \to t_2}_{p_A} \% N \tag{11}$$

$$C^{t_l \to t_2}_{p_A} = \begin{cases} 0, & if\ X^{t_l}_{p_A} = X^{t_2}_{p_A} \\ 1, & if\ X^{t_l}_{p_A} \neq X^{t_2}_{p_A} \end{cases} \tag{12}$$

where N denotes the number of land cover classes. Meanwhile, the ratios of source and target data may influence the performance of the model, which can be seen as a hyperparameter of our framework, and we will discuss in the ablation study.

IV. Experiments

A. Experiments Settings

1) Datasets: We select three open-source SCD datasets to evaluate our method UPchange. **SECOND dataset** [50]: The SECOND dataset is an SCD that contains 4662 pairs of aerial images, each with a size of 512×512 pixels and a resolution ranging from 0.3 to 5 m. It covers six classes: nonvegetated ground surface, tree, low vegetation, water, buildings, and playgrounds. However, its semantic annotations only cover the change areas, meaning that the semantic change labels will be sparse in this dataset when using the method we proposed. Therefore, we use the whole dataset to evaluate our method, specifically speaking, 2968 pairs of images for training and 1694 pairs of images for testing. **Hi-UCD mini dataset** [51]: The Hi-UCD mini dataset tries to discover the urban semantic changes in Tallinn, the capital of Estonia, including nine land cover classes: water, grass, building, greenhouse, road, bridge, bare land, woodland, and others. The mini dataset covers an area of 30 km^2, which contains 300 pairs of images for training and 386 pairs of images for testing, with each image having 1024×1024 pixels and each pixel covering a very high resolution of 0.1 m. **WUSU dataset** [52]: The WUSU dataset is a triple-temporal SCD dataset for two areas in Wuhan, China. It includes eleven land cover classes, which means a challenge for the SCD task. The WUSU dataset covers an area of nearly 80 km^2, with 6358×6382 pixels obtained for April 22, 2015, September 1, 2016, and November 23,

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2018 for Hongshan District and 7025 × 5500 pixels obtained for April 22, 2015, December 8, 2016, and November 23, 2018 for Jiang'an District. Unlike the original setting of the dataset with four-band fusion satellite images with 1-m spatial resolution, we chose the former three-band image like the other two datasets.

2) Metrics: We evaluate our method from two aspects: BCD and SCD. For BCD, we first use the overall accuracy (OA) to assess the overall performance of the model. Additionally, since OA will be dominated by negative samples because of the imbalance of change and unchange pixels in the change detection dataset, we adapt the intersection over union (IoU), which computes scores separately for each class

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{14}$$

where *TP, TN, FP*, and *FN* indicate true positive, true negative, false positive, and false negative pixels. For every class, we adopt F_1score as a metric, which is the harmonic mean of precision and recall

$$precision = \frac{TP}{TP + FP} \tag{15}$$

$$recall = \frac{TP}{TP + FN} \tag{16}$$

$$F_1 = 2 \times \frac{precision * recall}{precision + recall}. \tag{17}$$

For SCD, to evaluate the accuracy of every change type, we adopt separated Kappa (SeK) coefficient, which is based on the kappa coefficient; different from the Kappa coefficient assign same weights to every class, SeK assigns more weight to the change class which fit to SCD task

$$SeK = e^{IoU_1} \frac{\hat{\rho} - \hat{\eta}}{1 - \hat{\eta}} \tag{18}$$

$$\hat{\rho} = \frac{\sum_{i=2}^{C} M_{ii}}{\sum_{i=1}^{C} \sum_{j=1}^{C} M_{ij} - M_{11}} \tag{19}$$

$$\hat{\eta} = \frac{\sum_{j=1}^{C} \left(\hat{M_{i+}} \hat{M_{+j}}\right)}{\left(\sum_{i=1}^{C} \sum_{ij=1}^{C} M_{ij} - M_{11}\right)^2} \tag{20}$$

in which we set the SCD metric is denote as $M \in R^{n*n}$, where n is the number of semantic change class, M_{11} denotes the number of true positive pixels of unchanged class, $\hat{M_{i+}}, \hat{M_{+j}}$ denote the sum of over row and column without M_{11} on confusion matrix, respectively.

3) Implementation Details: To ensure a fair comparison between the proposed method and other methods, we trained all models with a total batch size of 16 for specific iterations. Specifically, for the SECOND dataset, we train it for a total of 25k iterations. The first 10 000 iterations were used for warming up with only source labels, and the subsequent iterations used both source labels and pseudo-labels generated every 5k iterations. As the Hi-UCD mini dataset, the training process is conducted for 40k iterations, with the first 10k iterations for warming up and the next 30k iterations for training with both source and pseudo-labels. The Hi-UCD mini dataset requires more computation to converge due to its larger number of land cover classes.

We randomly cropped the images into 256 × 256 pixel patches, and the self-paired data were added every ten steps. Data augmentation techniques, such as horizontal and vertical flipping, random rotation, and random color jitter, were applied during training. The stochastic gradient descent(SGD) method was chosen to optimize the model with a weight decay of 0.0001 and a momentum of 0.09. The initial learning rate was set to 0.03, and a "poly" learning rate policy with a decay factor of 0.9 was applied. We trained our semantic change detector on an NVIDIA GeForce RTX 3090 GPU, with a memory usage of approximately 20 GB.

B. Comparison Experiments

1) Benchmark Methods: To fairly evaluate the proposed method, we choose the following methods to conduct experiments under the same training protocol. These methods can be categorized into three fields: single-temporal supervised SCD, single-temporal supervised SCD with bitemporal unsupervised domain adaptation, and bitemporal supervised SCD.

For single-temporal supervised SCD, we chose some single-temporal supervised methods and change-synthesized methods.

1) *HRSCD Strategy 1 [9]:* PCC is an intuitive way for SCD, which tries to train a land cover classification model at first and then compares the land cover results at different times. So, PCC can be trained on unpaired data without any temporal information.
2) *ChangeStar [19]:* ChangeStar is a simple yet effective multitask change detection architecture. This work proposed a single-temporal supervised learning (STAR) algorithm, enabling the change detector to be trained with unpaired data, i.e., single-temporal data. The STAR algorithm breaks the restriction of change detection models that need paired images and dense labels.
3) *ChangeStar2 [20]:* Compared to ChangeStar, ChangeStar2 introduces improvements in both network architecture and training strategy. Specifically, it incorporates a temporal difference network to mitigate the issue of slow convergence. Additionally, self-contrast samples are introduced during the training process to increase the proportion of unchanged labels.
4) *Changen2 [35]:* Change2 is a generative change detection framework that achieves the generation of change detection data by introducing editable semantic masks as conditional inputs into a diffusion model. In this way, it can leverage single-temporal data and labels to generate a large number of postchange images, achieving change detection by single-temporal data from a different perspective.
5) *HySCDG [36]:* HySCDG proposes a generative pipeline for creating a large hybrid SCD dataset that contains both real VHR images and inpainted ones, along with a land cover semantic map at both dates and the change map.

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TABLE I

COMPARISON RESULTS ON THE SECOND DATASET,BOLD FONT INDICATE THE BEST INDEX UNDER THE SETTING OF SINGLE-TEMPORAL SEMANTIC SUPERVISED LEARNING.

Methods	Binary Change Detection			Semantic Change Detection		Requirement
	$OA(\%)$	$IoU(\%)$	$F_1(\%)$	$Kappa(\%)$	$SeK(\%)$	
Bi-temporal semantic supervised learning						
HRSCD strategy4	87.28	70.04	81.47	27.89	16.80	pre(img,label),post(img,label)
ChangeMask	89.14	70.97	81.91	30.37	18.25	pre(img,label),post(img,label)
Single-temporal semantic supervised learning without UDA						
HRSCD strategy1	72.52	51.97	67.04	16.92	8.98	pre(img,label)
ChangeStar	72.06	51.25	66.32	15.01	7.88	pre(img,label)
ChangeStar2	78.15	56.59	70.55	15.29	8.27	pre(img,label)
Changen2	85.99	62.45	75.60	16.29	9.41	pre(img,label)
HySCDG	86.45	63.26	77.28	17.26	10.04	pre(img,label)
UPchange w.o TCA	78.21	57.62	71.66	19.00	10.52	pre(img,label)
Single-temporal semantic supervised learning with UDA						
+DAN	$85.24_{(+7.03)}$	$63.82_{(+6.20)}$	$75.85_{(+4.19)}$	$19.79_{(+0.79)}$	$10.53_{(+0.01)}$	pre(img,label),post(img)
+AdaptSeg	$72.42_{(-5.79)}$	$52.10_{(-5.52)}$	$67.23_{(-4.43)}$	$17.41_{(-1.59)}$	$9.29_{(-1.23)}$	pre(img,label),post(img)
+ADVEnt	$85.47_{(+7.26)}$	$60.60_{(+2.98)}$	$72.74_{(+1.08)}$	$15.38_{(-1.62)}$	$8.20_{(-2.32)}$	pre(img,label),post(img)
+FDA	$79.61_{(+1.40)}$	$59.17_{(+1.55)}$	$71.42_{(-0.12)}$	$19.15_{(+0.15)}$	$10.35_{(-0.17)}$	pre(img,label),post(img)
+GLGAN	$80.96_{(+2.75)}$	$51.32_{(-6.3)}$	$67.42_{(-4.24)}$	$14.43_{(-4.57)}$	$7.35_{(-2.88)}$	pre(img,label),post(img)
+CBST	$77.81_{(-0.4)}$	$57.69_{(+0.07)}$	$71.86_{(+0.20)}$	$20.28_{(+1.28)}$	$11.78_{(+1.26)}$	pre(img,label),post(img)
UPchange	$86.07_{(+7.86)}$	$\mathbf{63.38}_{(+5.76)}$	$\mathbf{75.51}_{(+3.85)}$	$\mathbf{21.77}_{(+2.77)}$	$\mathbf{12.22}_{(+1.70)}$	pre(img,label),post(img)

For bitemporal SCD, UPchange can use an unsupervised domain adaptation method as a weak supervision signal, though it cannot be seen as a strong restriction as a common bitemporal supervised method. We also chose a state-of-the-art bitemporal supervised semantic detection framework to prove the effectiveness and robustness of our method.

1) *HRSCD Strategy 4 [9]:* HRSCD strategy4 is the first method focus on SCD. It is a Siamese multitask framework that decouples the SCD task into a change detection task and a semantic segmentation task. This method was based on pairwise data and labels, so a multitemporal dataset was needed.
2) *ChangeMask [26]:* ChangeMask is the state-of-the-art SCD method that simply the original Siamese multitask framework by removing the change encoder branch. Meanwhile, it adds a temporal symmetric fusion module that can avoid overfitting the time order.

Meanwhile, an SCD framework that can be trained on unpaired data with only semantic labels was proposed; the model trained on single-temporal data cannot fit on real application which based on bitemporal data. This is because of the domain shift, so we propose a UDA method based on CBST to eliminate the difference; meanwhile, any other UDA methods can be used on this framework, so we choose some UDA methods as our benchmark methods.

1) *Deep Adaptation Network [53]:* DAN is a discrepancy-based deep domain adaptation method, which can learn transferable features with statistical guarantees, and can scale linearly by an unbiased estimate of kernel embedding.
2) *AdaptSegNet [43]:* AdaptSegNet is an adversarial method that sets a zero-sum game on the feature level output of the source and target domains.
3) *ADVENT [44]:* As the AdaptSegNet mentioned above, ADVENT is also an adversarial method; however, the antagonism doesn't act on the feature level, but on the entropy of the output.
4) *CBST [48]:* CBST is a self-training UDA method that trains a model with labels on the source domain and pseudo-labels on the target domain. The selection of the target domain was based on the confidence of each pixel.
5) *Fourier Domain Adaptation [54]:* FDA is a Fourier transform-based domain adaptation method that efficiently generates intermediate data, which incorporates the semantics of the source domain and the style of the target domain, by transforming source and target domain images into the frequency domain using the Fourier transform, thereby achieving domain adaptation.
6) *GLGAN [55]:* GLGAN is an adversarial training-based method that enhances domain fusion by extracting multiscale image features and incorporating a global–local adversarial network.

As mentioned above, the UPchange framework can train a semantic change detector using only single-temporal unpaired data with semantic annotations. Therefore, we conducted experiments with some classic single-temporal SCD methods on two open-source datasets. Since bitemporal images can be used as unsupervised domain adaptation data, to explore the upper limit of our work, we chose two state-of-the-art SCD methods based on bitemporal supervised learning: HRSCD strategy4 and ChangeMask. In order to get the maximum extent of the fair, the original settings of the train/test/val set were used, and, in our method, we shuffle the paired data

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TABLE II

COMPARISON RESULTS ON THE HI-UCD MINI DATASET,BOLD FONT INDICATE THE BEST INDEX UNDER THE SETTING OF SINGLE-TEMPORAL SEMANTIC SUPERVISED LEARNING.

Methods	Binary Change Detection			Semantic Change Detection		Requirement
	$OA(\%)$	$IoU(\%)$	$F_1(\%)$	$Kappa(\%)$	$SeK(\%)$	
Bi-temporal semantic supervised learning						
HRSCD strategy4	93.92	72.42	82.24	22.73	13.97	pre(img,label),post(img,label)
ChangeMask	95.49	77.56	86.25	34.37	23.03	pre(img,label),post(img,label)
Single-temporal semantic supervised learning without UDA						
HRSCD strategy1	88.93	64.35	75.73	19.29	10.66	pre(img,label)
ChangeStar	88.72	64.31	75.75	13.78	7.63	pre(img,label)
ChangeStar2	91.18	68.35	79.12	22.44	13.11	pre(img,label)
Changen2	91.96	69.52	79.60	15.34	9.31	pre(img,label)
HySCDG	94.00	73.21	80.90	16.40	9.17	pre(img,label)
UPchange w.o TCA	90.69	66.98	77.91	21.48	12.27	pre(img,label)
Single-temporal semantic supervised learning with UDA						
+DAN	$94.20_{(+3.51)}$	$71.48_{(+4.50)}$	$81.35_{(+3.44}$	$28.48_{(+7.00)}$	$17.13_{(+4.86)}$	pre(img,label),post(img)
+AdaptSeg	$93.50_{(+2.81)}$	$69.59_{(+3.61)}$	$79.76_{(+1.85)}$	$23.49_{(+2.01)}$	$13.70_{(+1.03)}$	pre(img,label),post(img)
+ADVEnt	$93.86_{(+3.17)}$	$70.55_{(+3.57)}$	$80.57_{(+2.66)}$	$22.92_{(+1.44)}$	$13.57_{(+1.30)}$	pre(img,label),post(img)
+FDA	$90.71_{(+0.02)}$	$66.6_{(-0.38)}$	$77.53_{(-0.38)}$	$18.85_{(-2.63)}$	$10.68_{(-1.59)}$	pre(img,label),post(img)
+GLGAN	$87.88_{(-2.81)}$	$61.91_{(-5.07)}$	$75.11_{(-2.80)}$	$23.39_{(+1.91)}$	$11.50_{(-0.77)}$	pre(img,label),post(img)
+CBST	$94.61_{(+3.93)}$	$73.86_{(+3.88)}$	$83.36_{(+5.45)}$	$30.40_{(+8.98)}$	$19.09_{(+6.82)}$	pre(img,label),post(img)
UPchange	$\mathbf{94.84}_{(+4.15)}$	$\mathbf{74.76}_{(+7.78)}$	$\mathbf{84.08}_{(+6.17)}$	$\mathbf{32.92}_{(+11.44)}$	$\mathbf{21.00}_{(+8.73)}$	pre(img,label),post(img)

in the training set to eliminate temporal information during semantic supervised learning. All of these methods used the same deep learning model, EfficientNet-b0, as the semantic encoder backbone.

Table I shows the comparison results on the SECOND dataset, three phenomena can be found. First, UPchange outperforms other STAR methods, with an increase of 4%–6% in BCD and 1%–2% in SCD. Second, the unsupervised domain adaptation methods we proposed enhance the performance of the model, with every index in both BCD and SCD showing an improvement of 1%–8% compared to the STAR method without domain adaptation. However, other unsupervised domain adaptation methods fail to produce positive results on this dataset, which may be caused by the inadequacy of the semantic annotations of this dataset. Besides, we find that the self-training-based adaptation methods can achieve better performance than others, which means these methods are more suitable for our framework. And, we find that change-synthesized single-temporal methods achieved impressive results in BCD, surpassing our method on this dataset. However, these approaches perform poorly on SCD tasks. The reason may be that the differences between the generated data and the test set introduce noise interference in the learning of semantic representations, leading to a decline in the classification branch's accuracy.

Table II presents the comparison results on the Hi-UCD mini dataset. The first phenomenon is similar to the SECOND dataset, where our method achieved the best performance in the STAR setting without domain adaptation. Unlike the SECOND dataset, we observe that all UDA methods outperform the STAR method without domain adaptation. Furthermore, the proposed method achieves the highest performance among all UDA methods. An even more exciting finding is that, thanks to the diversity of change label—albeit pseudo-labels—our method can even outperform the bitemporal paired supervised learning methods, HRSCD strategy4, despite lacking temporal consistency information.

Table III presents the comparative experimental results on the WUSU dataset. Although it is a challenging dataset, our method still achieves the best performance among both single-temporal and domain adaptation approaches. In the single-temporal comparison, our method improves the change detection performance by 1%. In the domain adaptation comparison, it enhances change detection by 11% and SCD by 1.5%. Similarly, it can be observed that in this dataset, self-training-based domain adaptation methods can achieve competitive results, which proves the argument presented earlier.As the change-synthesized methods, we find these methods are ineffective on this dataset; this may be due to a severe mismatch between the land cover types defined in the generated data and those specified in the dataset.

The visualization results of the three datasets were shown at Figs. 5–7. We can observe that the results from STAR contain a lot of false alarms; these false alarms can be effectively eliminated by incorporating unlabeled bitemporal images. UPchange effectively eliminates most of these false alarms compared to other methods in this series, and its visual results are even comparable to those of bitemporal approaches. In terms of semantics, our method also more closely approximates the results of bitemporal supervised learning approaches, exhibiting smaller visual discrepancies from the ground truth.

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TABLE III

COMPARISON RESULTS ON THE WUSU DATASET,BOLD FONT INDICATE THE BEST INDEX UNDER THE SETTING OF SINGLE-TEMPORAL SEMANTIC SUPERVISED LEARNING.

Methods	Binary Change Detection			Semantic Change Detection		Requirement
	$OA(\%)$	$IoU(\%)$	$F_1(\%)$	$Kappa(\%)$	$SeK(\%)$	
Bi-temporal semantic supervised learning						
HRSCD strategy4	93.79	63.27	73.13	8.01	4.09	pre(img,label),post(img,label)
ChangeMask	94.19	65.4	75.36	10.77	5.72	pre(img,label),post(img,label)
Single-temporal semantic supervised learning without UDA						
HRSCD strategy1	77.97	49.52	61.75	7.41	3.42	pre(img,label)
ChangeStar	68.84	41.94	54.68	3.89	1.69	pre(img,label)
ChangeStar2	74.30	46.47	59.01	5.95	2.69	pre(img,label)
Changen2	80.24	49.77	58.23	N/A	N/A	pre(img,label)
HySCDG	85.71	52.65	59.85	N/A	N/A	pre(img,label)
UPchange w.o TCA	80.42	51.51	**63.46**$_{(+1.71)}$	**7.68**$_{(+0.27)}$	**3.59**$_{(+0.17)}$	pre(img,label)
Single-temporal semantic supervised learning with UDA						
+DAN	87.69$_{(+7.27)}$	52.30$_{(+0.79)}$	63.56$_{(+0.10)}$	7.84$_{(+0.18)}$	3.69$_{(+0.10)}$	pre(img,label),post(img)
+AdaptSeg	92.59$_{(+12.17)}$	55.95$_{(+4.44)}$	64.31$_{(+0.85)}$	3.39$_{(-4.29)}$	1.62$_{(-1.97)}$	pre(img,label),post(img)
+ADVEnt	92.82$_{(+12.40)}$	60.93$_{(+9.42)}$	70.72$_{(+7.26)}$	3.72$_{(-3.96)}$	1.85$_{(-1.74)}$	pre(img,label),post(img)
+FDA	79.95$_{(-0.47)}$	51.02$_{(-0.49)}$	62.99$_{(-0.47)}$	4.96$_{(-2.72)}$	2.31$_{(-1.28)}$	pre(img,label),post(img)
+GLGAN	79.30$_{(-1.12)}$	52.99$_{(+1.48)}$	62.43$_{(-1.03)}$	3.85$_{(-3.83)}$	2.19$_{(-1.40)}$	pre(img,label),post(img)
+CBST	89.54$_{(+9.12)}$	61.48$_{(+9.97)}$	72.45$_{(+8.99)}$	8.15$_{(+0.47)}$	4.21$_{(+0.62)}$	pre(img,label),post(img)
UPchange	91.30$_{(+10.88)}$	**64.52**$_{(+13.01)}$	**75.25**$_{(+11.79)}$	**9.38**$_{(+1.70)}$	**5.06**$_{(+1.47)}$	pre(img,label),post(img)

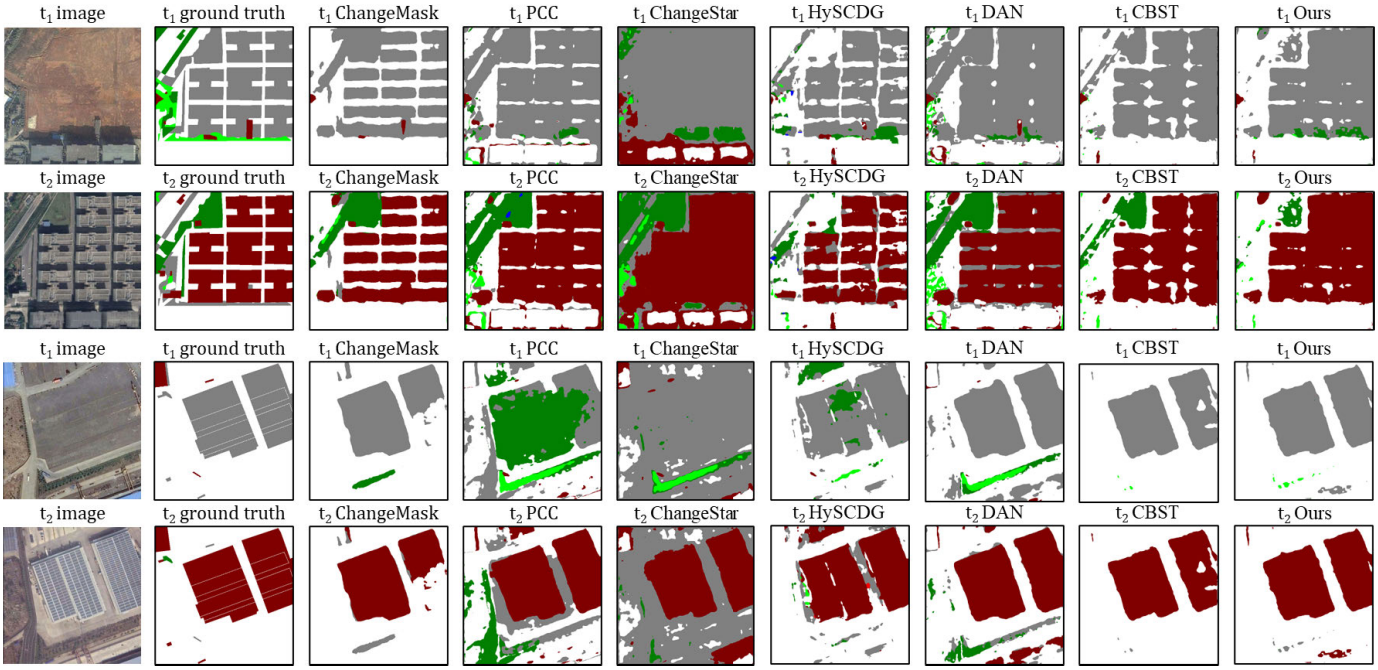


Fig. 5. Visualization of SCD results of compared methods on the SECOND dataset. Legends: water, nonvegetated ground, low vegetation, tree, buildings, playgrounds.

C. Ablation Study

Our method incorporates several improvements to address the lack of temporal information in unpaired data: 1) formulation on types of change; 2) self-pair contrast; and 3) a dynamic self-training unsupervised domain adaptation strategy with mixed percentage and threshold. To further analyze the mechanism of the proposed framework, a comprehensive ablation study was conducted on the Hi-UCD mini dataset and the SECOND dataset to evaluate the specific improvement by the key components. Since UPchange is based on the formulation of types of change, we first compare this formulation to other unpaired methods and then use it as our benchmark for other ablation experiments. We selected the overall accuracy(OA) and mF_1 score of BCD and separated the Kappa (SeK) coefficient of SCD as metrics.

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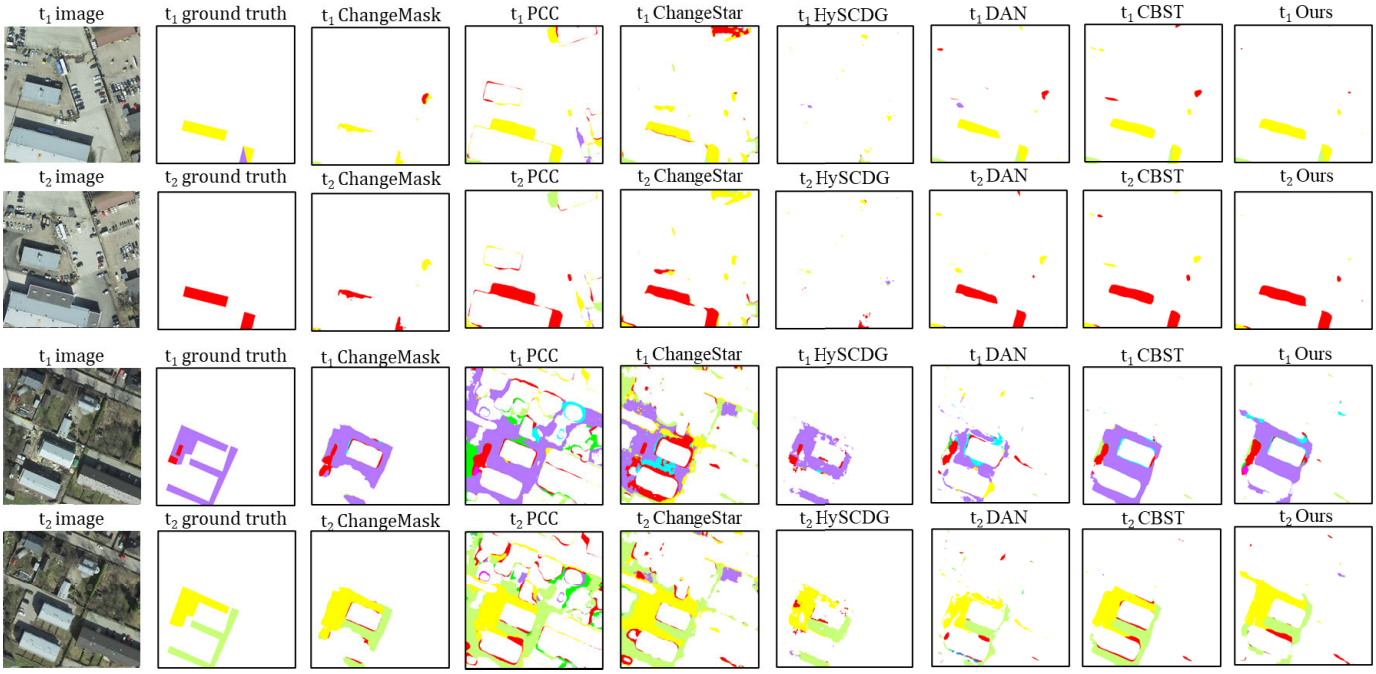


Fig. 6. Visualization of SCD results of compared methods on Hi-UCD mini dataset. Legends: lake, grass, building, greenhouse, road, bride, tree, bare land, and others.

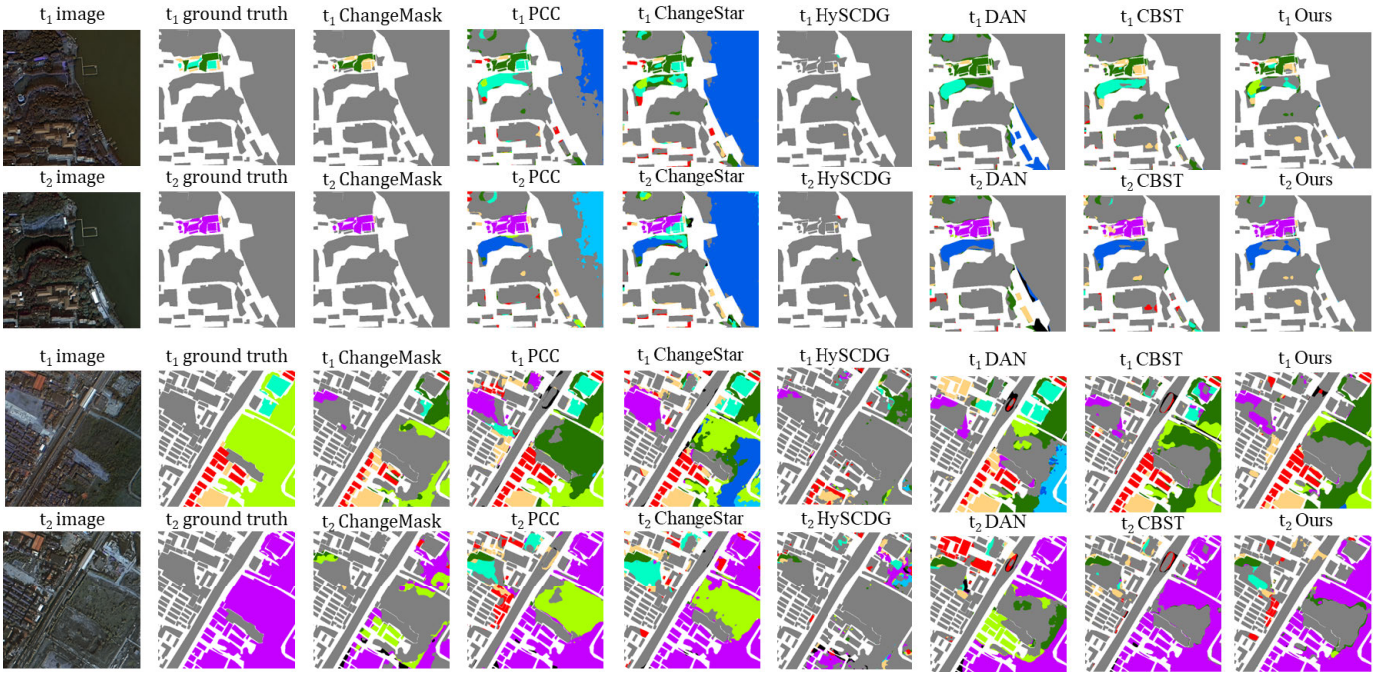


Fig. 7. Visualization of SCD results of compared methods on the WUSU dataset. Legends: unclassified, road, low building, high building, arable land, woodland, grassland, river, lake, structure, excavation, bare surface.

1) Formulation on Types of Change: The core idea of our method is the formulation of types of change based on unpaired data with semantic annotations. We conducted comprehensive experiments on STAR using the SECOND and Hi-UCD mini datasets. Different from the original setup, which splits the train and test sets with paired bitemporal images and semantic annotations, during the process of training, we only use the image and semantic annotations from the first phase. We use ResNet as our semantic-aware encoder. ResNet [56] proposes a residual architecture by adding an identity mapping to solve the degradation problem. To explore the influence of the model's scale, we experiment with four different ResNet encoder backbones of varying scales on the SECOND and Hi-UCD mini datasets.

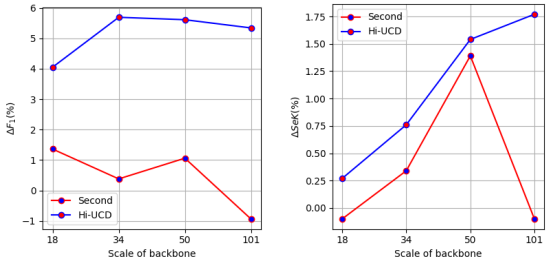
Table IV shows the comparison results on the SECOND and Hi-UCD mini dataset. It can be observed that even without

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TABLE IV

SINGLE-TEMPORAL SEMANTIC SUPERVISED LEARNING RESULTS ON THE SECOND AND HI-UCD MINI DATASET, BOLD FONT INDICATE THE BEST INDEX UNDER EVERY SEMANTIC-AWARE ENCODER BACKBONE

Methods	Second				Hi-UCD mini			
	BCD		SCD		BCD		SCD	
	$IoU(\%)$	$F_1(\%)$	$Kappa(\%)$	$SeK(\%)$	$IoU(\%)$	$F_1(\%)$	$Kappa(\%)$	$SeK(\%)$
ResNet-18								
HRSCD strategy1	45.62	61.32	11.22	5.65	56.21	68.53	10.13	5.08
ChangeStar	53.33	67.78	9.49	4.98	53.32	65.96	7.03	3.43
UPchange w.o TCA	**55.02**	**69.14**	10.45	5.55	**60.97**	**72.58**	**10.19**	**5.35**
ResNet-34								
HRSCD strategy1	44.98	60.69	9.79	4.90	55.02	67.47	9.77	4.84
ChangeStar	52.98	67.47	9.32	4.87	52.52	65.38	8.38	4.08
UPchange w.o TCA	**53.46**	**67.85**	**9.99**	**5.24**	**61.59**	**73.16**	**10.57**	**5.60**
ResNet-50								
HRSCD strategy1	44.96	60.74	10.03	5.03	55.09	67.60	9.74	4.84
ChangeStar	52.92	67.48	9.48	4.97	57.25	69.57	10.22	5.20
UPchange w.o TCA	**54.02**	**68.54**	**12.05**	**6.42**	**64.05**	**75.18**	**12.41**	**6.74**
ResNet-101								
HRSCD strategy1	45.89	61.57	10.85	5.47	56.01	68.34	9.71	4.86
ChangeStar	53.35	67.83	9.14	4.81	55.46	68.08	10.55	5.29
UPchange w.o TCA	52.18	66.89	10.29	5.37	**62.19**	**73.87**	**13.12**	**7.06**



(a) (b)

Fig. 8. Increase of our method compared to the best index of other single-temporal methods, △ indicates the accuracy gap between two methods on different backbones. (a) Increase on F_1. (b) Increase on SeK.

any domain adaptation methods to eliminate the domain gap between unpaired and paired data, our method performs better than other single-temporal unpaired SCD methods in both BCD and SCD metrics across almost all semantic encoder backbones. Specifically speaking, UPchange achieves an average increase of 0.5% in F_1 score for BCD and 0.4% in SeK on the SECOND dataset; the improvements in these two indices are about 5.2% and 1.1% on the Hi-UCD mini dataset.

Overall, comparing the results of the two datasets, it can be found that our method gets better performance on the Hi-UCD mini dataset in the BCD metric, even though with a larger change type space. As Fig. 8 shows, the increase on Hi-UCD mini is better than on SECOND across different backbones. Additionally, the F_1 score and Sek even decrease when deeper models were used. This suggests that sufficient semantic labels are beneficial to the learning of semantic change in our method.

TABLE V

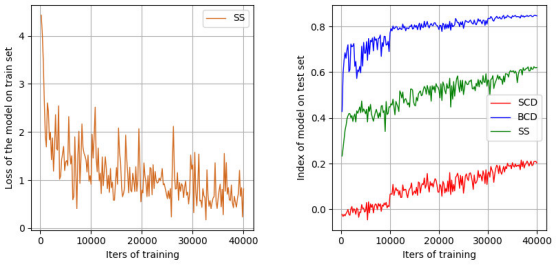
ABLATION EXPERIMENTS RESULTS ON THE HI-UCD MINI DATASET TO UNDERSTAND THE CONTRIBUTION OF EACH COMPONENT

Method	Formulation	Self-pair	UDA	$OA(\%)$	$mF_1(\%)$	$SeK(\%)$
(a)ChangeType	✓			90.68	77.98	12.06
(b)+ Self-pair	✓	✓		90.69	77.91	12.27
(c)+ TCA	✓		✓	91.56	80.08	15.10
(d)UPchange	✓	✓	✓	94.84	84.08	21.00

2) Effectiveness of Self-Pair: To balance the large number of positive samples and the sparse negative samples, some self-pair samples were added during the training process. Table V—(a)–(d) shows the quantitative improvement of self-pair on baseline and baseline with UDA. Though the improvement in (a) and (b) is tiny, self-pair enhances the baseline with UDA by 3.28% OA and 4% F_1 for BCD and by 5.9% in SeK for SCD. This indicates that self-pair helps the model learn a more robust bitemporal semantic change representation, which is useful in the process of unsupervised domain adaptation.

3) Effectiveness of Bitemporal Domain Adaptation: The bitemporal unsupervised domain adaptation method aims to eliminate the domain gap between the single-temporal unpaired change space and the bitemporal paired space. Table V—(a)–(d) shows that using the domain adaptation

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(a) (b)

Fig. 9. Learning curves on Hi-UCD mini dataset. (a) Loss function curve. (b) Metric curve.

method we proposed results in a 2.1% increase in F_1 for BCD and a 3.04% increase in SeK for SCD over the baseline, as well as a 6.17% increase in F_1 for BCD and an 8.73% increase in SeK for SCD. These considerable improvements confirm the existence of a domain gap between bitemporal and single-temporal data.

To validate the effectiveness of the temporal change adaptation module, we plotted the convergence curve of the model's loss function and the performance metrics on the test set as Fig. 9. Using 200 training iters as an interval, we visualized the model's training loss, BCD metric (F1), land cover classification metric (F1), and SCD metric (SeK). The results indicate that after introducing unlabeled bitemporal data (at 10 000 training iterations), the loss curve exhibited a slight increase, demonstrating that the bitemporal data introduced a perturbation to the training process. Meanwhile, the performance curves on the test set revealed significant improvements in both BCD and SCD metrics after incorporating the bitemporal data, thereby confirming the effectiveness of the proposed module.

4) Importance of Semantic Supervision: In the methods section, we proposed that our framework has an optional semantic decoder branch, which may seem superfluous because the main branch of the framework already handles semantic information. However, due to the large space of the semantic change branch, the model is hard to converge, and a semantic branch helps the model learn a robust semantic representation. To demonstrate the importance of semantic supervision, we conducted supplementary experiments on the Hi-UCD mini dataset. In addition to the BCD and SCD metrics mentioned above, we also present quantitative results on semantic segmentation. Moreover, besides comparing our method with and without the semantic branch, we also show the HRSCD strategy 1's result, as it is a single-temporal strategy focused on the semantic segmentation.

Table VI shows the positive enhancement brought by semantic supervision. We can see that PCC performs well on semantic segmentation. However, due to the lack of temporal consistency constraints, it does not achieve satisfactory results in BCD and SCD. Our method without the semantic branch makes a breakthrough in predicting change areas, but because of the vast space of all change types, its performance is worse than PCC on semantic segmentation. Unlike the mutually exclusive effects of other multitask learning approaches, semantic representation learning enhances the performance of the SCD task. With the addition of semantic learning, our method achieves a 7.48% increase in the score for segmentation and a 2.79% improvement in SeK for SCD.

TABLE VI

SUPPLEMENT EXPERIMENT RESULTS ON THE HI-UCD MINI DATASET, BCD, SS, SCD MEAN THE RESULT OF BCD, SEMANTIC SEGMENTATION, AND SCD, RESPECTIVELY

	BCD		SS		SCD
Methods	$OA(\%)$	$mF_1(\%)$	$OA(\%)$	$mF_1(\%)$	$SeK(\%)$
PCC	88.93	75.73	84.96	61.16	10.66
w.o.semantic sup	94.65	84.63	82.95	55.31	18.21
w.semantic sup	94.84	84.08	84.62	62.79	21.00



Fig. 10. Visualization results of style transfer model on Hi-UCD dataset.

D. Discussion

1) Impact of Ratios on Source and Target Data: In the process of TCA, different ratios of the source and target data will influence the performance of the model. To explore the influence of the ratio on source and target data, we conducted a supplementary experiment on the Hi-UCD mini dataset. Different ratios between the source and target domain data were used to explore the impact of this parameter on model performance. Specifically, we conducted experiments using ratios of 0 (all source domain data), 0.2, 0.4, 0.6, 0.8, and 1 (all target domain data) with an interval of 0.2. The experimental results are presented in Table VII. The results indicate that the model achieves optimal performance when the volumes of source and target domain data are approximately balanced. However, the model's performance degrades significantly as the proportion of target domain data increases further (at ratios of 0.8 and 1.0). This suggests that relying solely on pseudo-labels leads to catastrophic predictions, likely due to the accumulation of errors from initial wrong pseudo-labels.

2) Impact of Domain Gap Between Bitemporal Images: Since the generated pseudo-change detection data lack temporal constraints, they fail to consider the domain gap between

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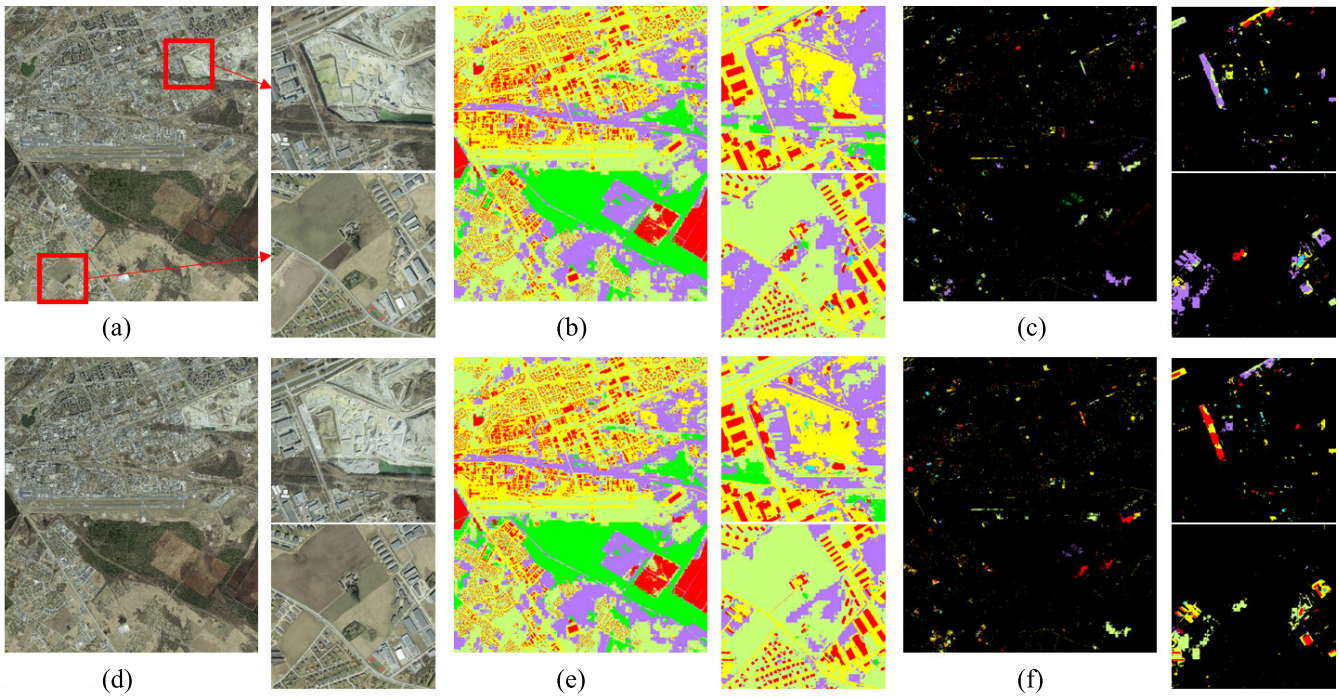


Fig. 11. Visualization of SCD inference results in Tallinn, the capital of Estonia. Legends: lake, grass, building, greenhouse, road, bride, tree, bare land, others. (a) 2018 image (60 000 × 70 000). (b) 2018 classification map. (c) 2018 semantic change map. (d) 2019 image (60 000 × 70 000). (e) 2019 classification map. (f) 2019 semantic change map.

TABLE VII
SUPPLEMENT EXPERIMENTS RESULTS ON THE HI-UCD MINI DATASET OF DIFFERENT RATIOS ON SOURCE AND TARGET DATA

Ratios	BCD		SS		SCD
	$OA(\%)$	$mF_1(\%)$	$OA(\%)$	$mF_1(\%)$	$SeK(\%)$
0	90.69	77.91	83.31	57.93	12.27
0.2	94.62	82.22	84.42	59.31	17.25
0.4	94.89	84.92	84.04	60.68	21.26
0.6	94.38	82.81	83.13	58.70	16.75
0.8	93.35	75.29	75.88	44.21	6.00
1	92.86	76.96	75.11	34.87	4.81

TABLE VIII
SUPPLEMENT STUDY FOR THE INFLUENCE OF DOMAIN GAP BETWEEN MULTITEMPORAL DATA

	BCD		SS		SCD
	$OA(\%)$	$mF_1(\%)$	$OA(\%)$	$mF_1(\%)$	$SeK(\%)$
Base	94.84	84.06	84.61	62.78	21.00
Unpaired	94.48	84.72	84.82	63.08	21.20
Pix2pixHD	94.83	85.30	85.27	64.79	22.41
MUNIT	94.96	85.47	85.25	64.69	22.28

bitemporal images. Although the TCA module alleviates this issue to some extent, it primarily addresses the domain gaps between bitemporal and single-temporal input data, rather than the domain gap between data across different times. To explore the influence of the domain gap between bitemporal images, we conducted supplementary experiments on the Hi-UCD dataset. First, we introduced temporal constraints to the input during supervise learning process, by inputting the images into the model with temporal order, while ensuring they were unpaired. Meanwhile, to maximize data utilization, we employed two style transfer methods, pix2pixHD [32] and MUNIT [57]. Pix2pixHD is a paired style transfer model, and MUNIT is an unpaired style transfer model; both these methods can eliminate the domain gap between bitemporal images by transferring the style of t1 image to t2 image and inversion. The result of these models can be seen in Fig. 10.

We use the results from the article as a baseline to explore model improvements with unpaired data and with the assistance of two style transfer methods, respectively. The results were shown in Table VIII. The results indicate that considering the bitemporal images' domain gap improves the model's performance to some extent. Specifically, the unpaired method, due to the suboptimal utilization of the data, shows slight improvement over the baseline model. In contrast, the generated data introduced by the style transfer methods—though not fully realistic in appearance—helps the model recognize the domain gap between bitemporal data, thereby enhancing the accuracy of the experiments. Although the experimental results show that incorporating temporal domain information can improve the model's performance, this actually imposes additional constraints during the supervised training process by providing some temporal information to the model.

E. Application

To demonstrate the inference and transfer capabilities of our method on large-scale semantic change maps, we use

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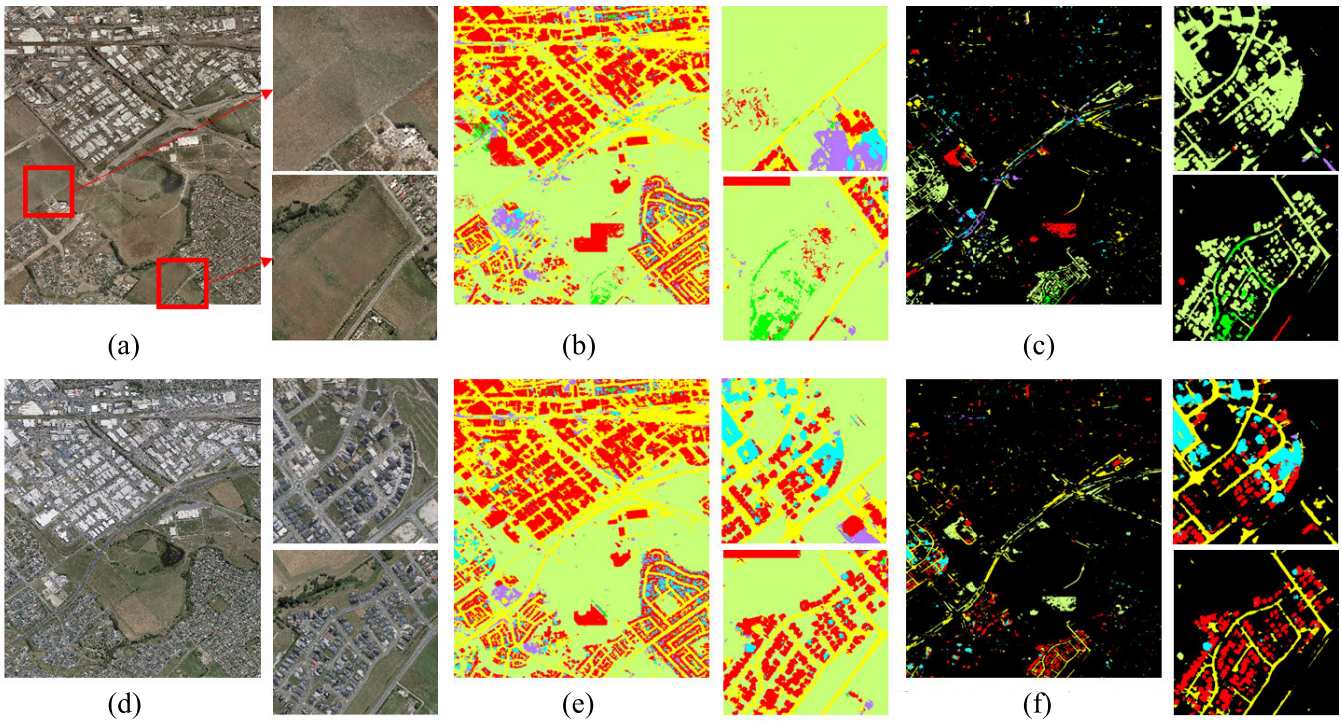


Fig. 12. Visualization of SCD inference results in Christchurch, New Zealand. Legends: lake, grass, building, greenhouse, road, bride, tree, bare land, others. (a) 2012 image (11 265 × 15 354). (b) 2012 classification map. (c) 2012 semantic change map. (d) 2016 image (11 265 × 15 354). (e) 2016 classification map. (f) 2016 semantic change map.

our model, trained on the Hi-UCD mini dataset, to infer the semantic change maps for two areas. The first area covers 42 km^2 with 4.2 billion pixels in Tallinn, the capital of Estonia, where the two temporal datasets were collected in 2018 and 2019. The second area corresponds to the training region of the WHU-Building dataset, covering an area of about 6.5 km^2 in Christchurch, New Zealand, with 173 million pixels, where a 6.3-magnitude earthquake occurred in February 2011 and subsequent rebuilding in the following years.

Fig. 11 shows the semantic map and semantic change map for a region in Tallinn, the capital of Estonia, in 2018 and 2019. We calculate the statistics of the results, approximately 85 million pixels changed during this period, covering about 0.85 km^2 of land, which accounts for about 2% of the area. The most frequent change was from grass to road, followed by bare land to grass, and grass to building; these three changes account for approximately 17.6%, 15.9%, and 12.2% of the change area, respectively. The statistics indicate that this region may have undergone rapid development during this period, as lots of man-made The statistics indicate that this region may have undergone rapid development during this period, as some man-made land features have replaced natural ones.

Fig. 12 shows the inference results of our model, trained on the Hi-UCD mini dataset for Christchurch, New Zealand, which experienced an earthquake in February 2011 and was captured in images from 2012 and 2016. Approximately 12 million pixels changed, covering about 7.2% of the area. The two most significant changes are the conversion of grass to roads and grass to buildings, each accounting for 21.1% of the total change area, indicating rapid recovery in this region after the earthquake. However, the third major change, the conversion of roads back to grass, accounts for approximately 13.9% of the change area, which could be attributed to false alarms caused by seasonal differences in the timing of the photographs.

V. Conclusion

In this work, we present a new SCD framework bypass the problem of existing methods that need a pairwise bitemporal dataset with annotations. To exploit potentialities of the existing land cover products, we proposed a new semantic change learning strategy based on single-temporal pseudo-semantic change labels and change type modeling. An unsupervised domain adaptation method was proposed to eliminate the domain transfer between single-temporal unpaired data and bitemporal paired data, which makes the model learn a robust semantic representation. The extensive experimental analysis shows its competitive performances in different domains with cheaper labels, which got the best performance in single-temporal methods, and even gets a better performance than some bitemporal supervised learning methods.As a limitation of our work, an excessively large semantic change space constrains both the convergence speed and effectiveness of the model. Since the method is based on semantic change types, which equals the square of the number of land-cover categories, this exponential expansion leads to a rapid growth in the semantic change space as the number of land-cover categories increases. In the future, limiting the number of

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semantic change types is a potential direction for model improvement. We hope our method will serve as a solid baseline and help ease future research.

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